# In-Game Chat Moderation and Censor

## 1. Project Description

This project is designed to build a system that could detect and censor in-game toxic words to ensure the game environments are friendly to every player. It is a neurosymbolic AI pipeline that could filter and moderate toxic behaviors in multiplayer gaming environments. This system employs a multistage NLP architecture that could process text and decide whether the texts are toxic and precisely masks only the offending tokens while preserving the player's original text.

## 2. NLP Task Description

The system pipeline is divided into three NLP tasks to balance computational efficiency and censorship accuracy.

### 1. Lexical Normalization

**NLP Task** : Seq2Seq/Text Transformation
**PIC** : Muhamad Nazih Najmudin (13523144)
**Description** :
It acts as the noise reduction layer. It receives the raw, unstructured in-game chat message and translates typos and gaming abbreviations into normalized English (e.g. "u r 1d1ot" to "u r idiot"). This task crucially generates a Character Index Map that aligns the normalized tokens back to the raw text, allowing downstream tasks to process clean data while ensuring the final censorship is applied to the original message.

### 2. Toxicity Classification

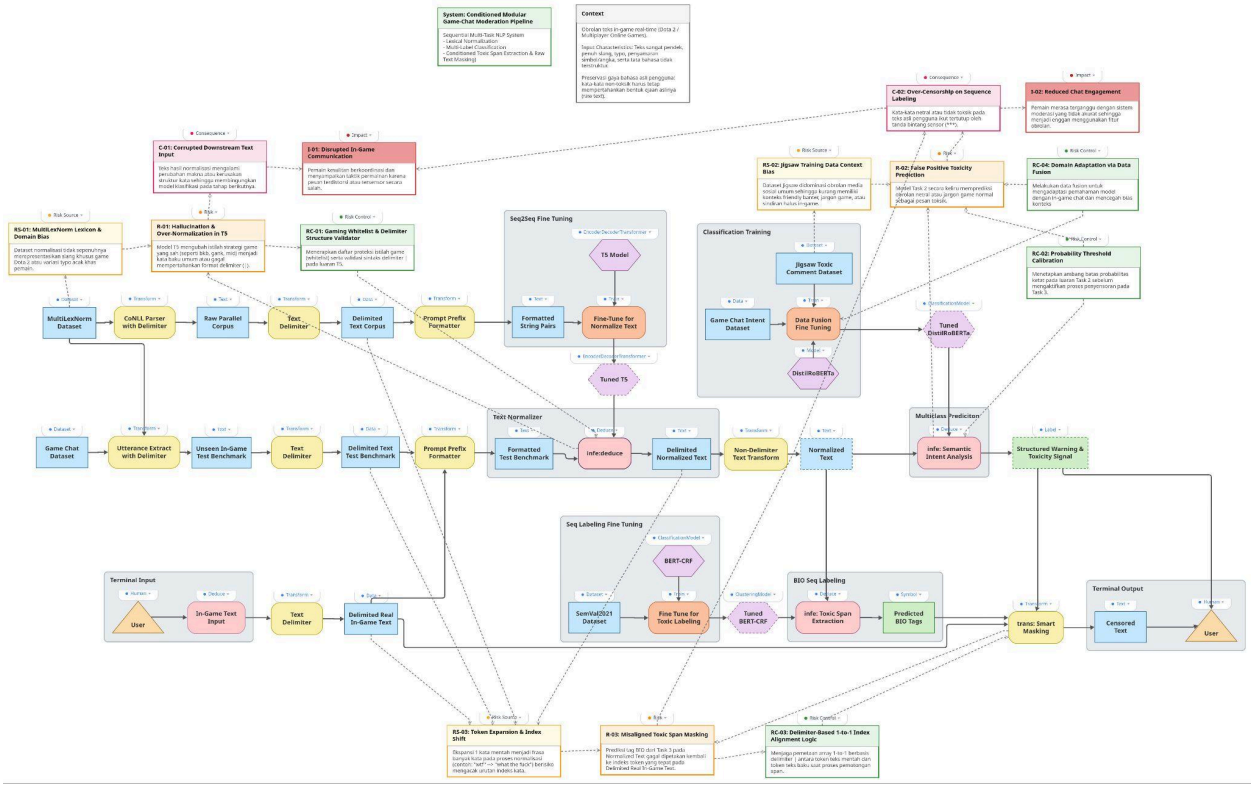
**NLP Task** : Text Classification
**PIC** : Ahmad Wafi Idzharulhaqq (13523131)
**Description** :
It performs multi label sequence classification on normalized in-game chat text using a fine tuned DistilRoBERTa model. By fusing general toxicity benchmarks with game chat intent dataset, this task acts as a semantic gatekeeper for evaluating player intent to dynamically trigger downstream token level extraction while simultaneously generating warning tags for toxicity if exist without suppressing the raw text.

### 3. Toxic Span Extraction & Censorship

**NLP Task** : Sequence Labeling
**PIC** : Ivant Samuel Silaban (13523129)
**Description** :
It acts as the token level execution layer. Receiving the delimited normalized text and the trigger signal from the classification phase, it utilizes a fine tuned BERT-CRF model trained on token-span datasets to

precisely identify and extract toxic character spans. Finally, it executes 1-to-1 index alignment to mask only the offensive segments on the original in-game message while preserving non toxic phrasing.

## 3. BEAM Notation




[BEAM Ontology Files]

## 4. System Explanation

This system is designed as a multi-stage neurosymbolic pipeline for in-game chat moderation in multiplayer games. The architecture divides processing into three natural language processing tasks to balance computational efficiency and censorship accuracy.

The first task is lexical normalization using sequence-to-sequence (seq2seq). It converts typos and gaming abbreviations into normalized English text while preserving original token structures using the | delimiter. We use the publicly available MultiLexNorm Dataset and raw parallel corpus to fine-tune the T5 model, alongside a team-built Game Chat Dataset serving as the test benchmark.

The second task is toxicity classification using a text classification approach. It analyzes normalized text from Task 1 to perform multilabel sequence classification (identifying categories such as insults, severe toxicity, or threats) using a fine-tuned DistilRoBERTa model. To bridge the domain gap, we employ Data

Fusion by combining the Jigsaw Toxic Comment Dataset with the Game Chat Intent Dataset, ensuring the model understands both general semantics and specific in-game contexts.

The third tasks are toxic span extraction and smart masking using sequence labeling like Named Entity Recognition (NER). It identifies specific positions of toxic words using a fine-tuned BERT-CRF model trained on the publicly available SemEval-2021 Dataset, followed by smart masking to map predicted BIO tags back to the raw user text.

There are three risk pathways. The first risk pathway originates from domain bias, where standard normalization datasets fail to capture game-specific slang and player typos. This bias induces T5 model hallucinations, causing valid gaming tactical terms to be incorrectly normalized or delimiter syntax to break. To mitigate this failure, we implemented a gaming whitelist and delimiter syntax validator to prevent downstream text from disrupting player communication.

The second risk stems from context bias in social media training data, which lacks friendly gaming banter and leads to false positive toxicity predictions. This risk is controlled through Data Fusion (RC-04) during training and probability threshold calibration (RC-02) applied to the classification outputs. Without these controls, over-censorship happens on neutral words, which ultimately reduces overall chat engagement.

The third pathway involves token expansion during normalization, where expanding single raw tokens into multi-word phrases causes index shifts. This shift creates misaligned toxic span masking, causing predicted tags to map back inaccurately to the original text. Implementing a delimiter-based 1-to-1 index alignment logic maintains proper token mapping, preventing over-censorship and communication breakdown.